# Conditional probability



## **1**: Set Notation

How sets are used to describe a sample space/event and how notation like $A \cap B$ is used to combine sets.

## **2**: Conditional Probability in Venn Diagrams

The notation $P(A|B)$ means "the probability of $A$ given that $B$ happened". How we can find such probabilities using a Venn Diagram.

## **3**: Formula for Conditional

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

## **4**: Tree Diagrams

"I have 3 red and 4 green balls in a bag. I take one ball out the bag, keep it, then take another. **Given that** the second ball was green, determine the probability the first was red."

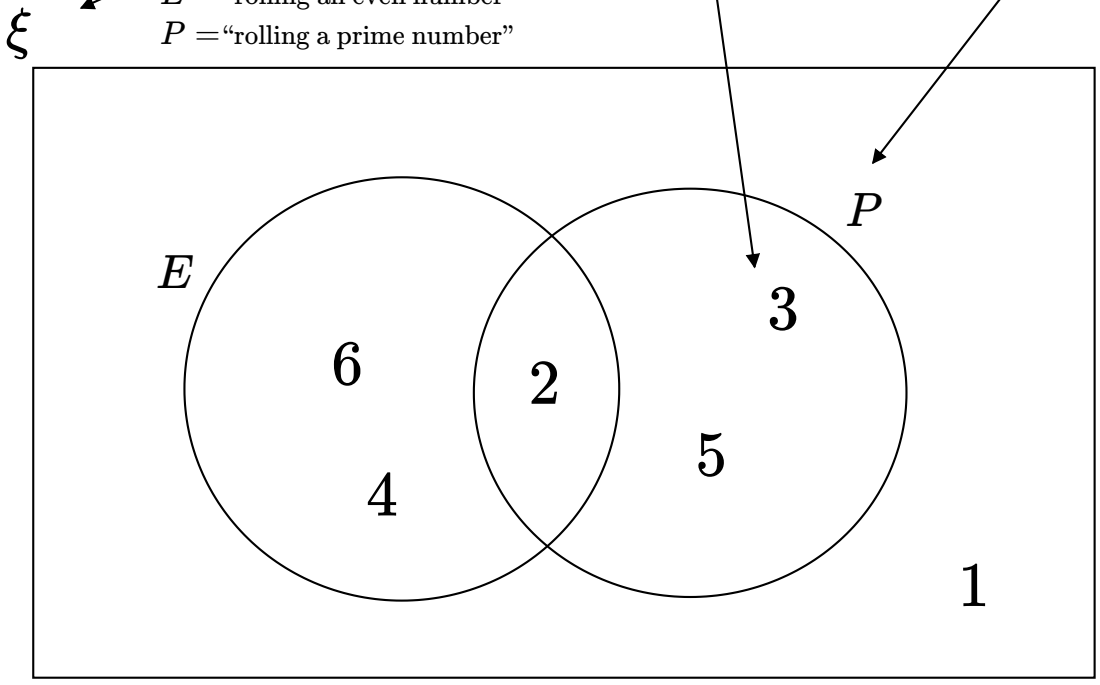
In general, sets are used to represent **collections of items**.

A **sample space** is **set of all possible outcomes**. We use $\xi$ (Greek 'Xi'), or sometimes just $S$, to represent this set. We use a rectangle in a Venn Diagram.

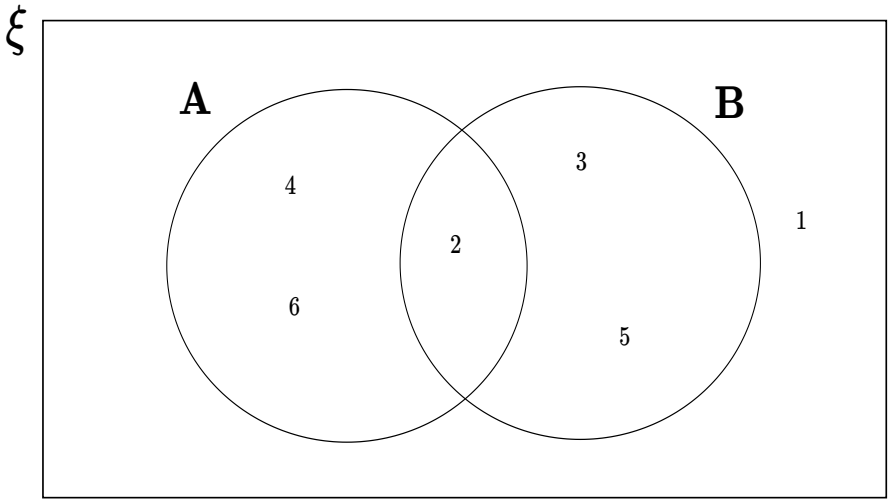
Each number represents an **outcome**.

In probability, an **event** is a **set of one or more outcomes**. These are the circles in the Venn Diagram. We use capital letters for the variables representing sets.

$E$ = "rolling an even number"
$P$ = "rolling a prime number"





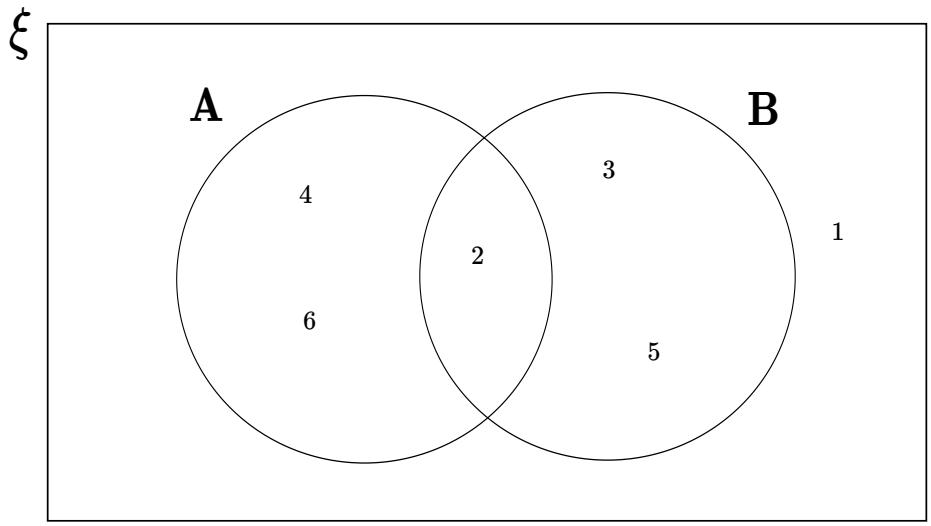




$\xi$ = the whole sample space (1 to 6)

$A$ = even number on a die thrown

$B$ = prime number on a die thrown

| | What does it mean in this context? | What is the resulting set of outcomes? |
|---|---|---|
| $A'$ | | |
| $A \cup B$ | | |
| $A \cap B$ | | |



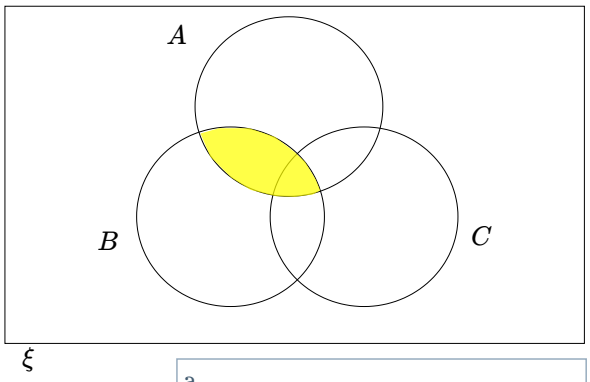

$S$ = the whole sample space (1 to 6)

$A$ = even number on a die thrown

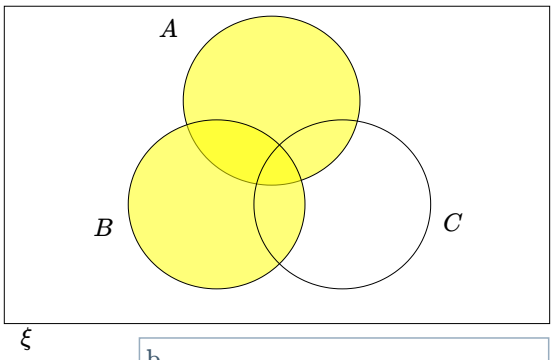
$B$ = prime number on a die thrown

| | What does it mean in this context? | What is the resulting set of outcomes? |
|---|---|---|
| $A \cap B'$ | | |
| $(A \cup B)'$ | | |
| $(A \cap B)'$ | | |

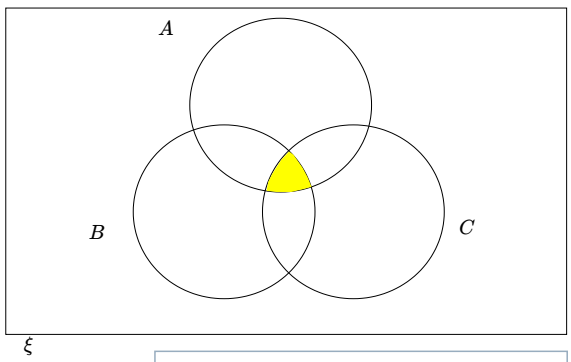




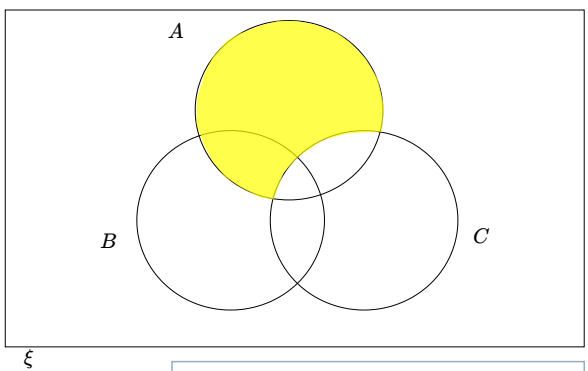

a




b

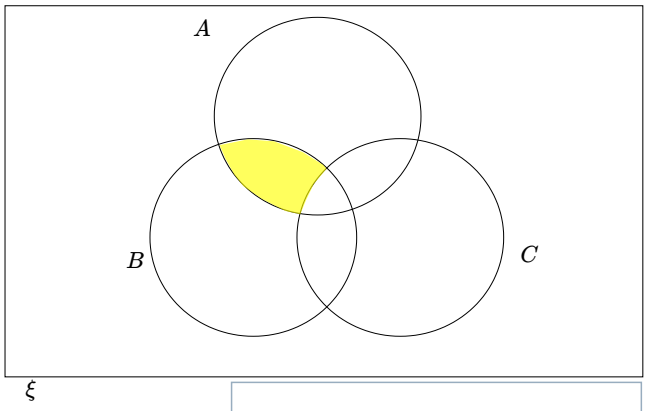


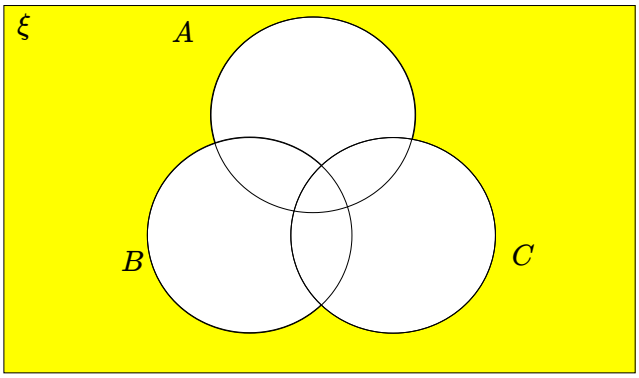

c



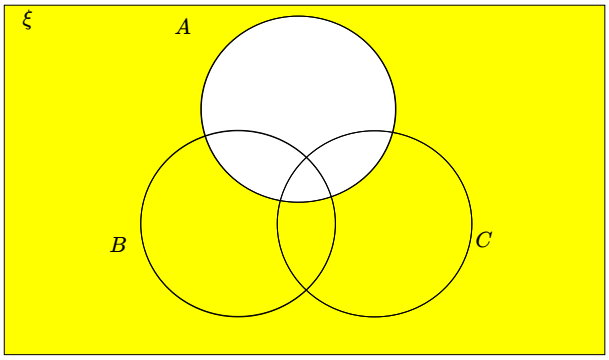

d

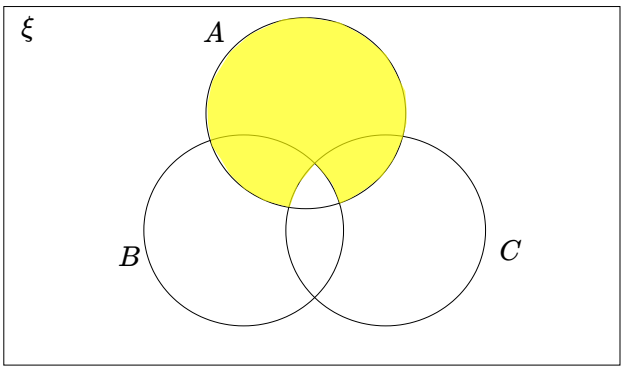
## What area is indicated? **Luisa · 7**



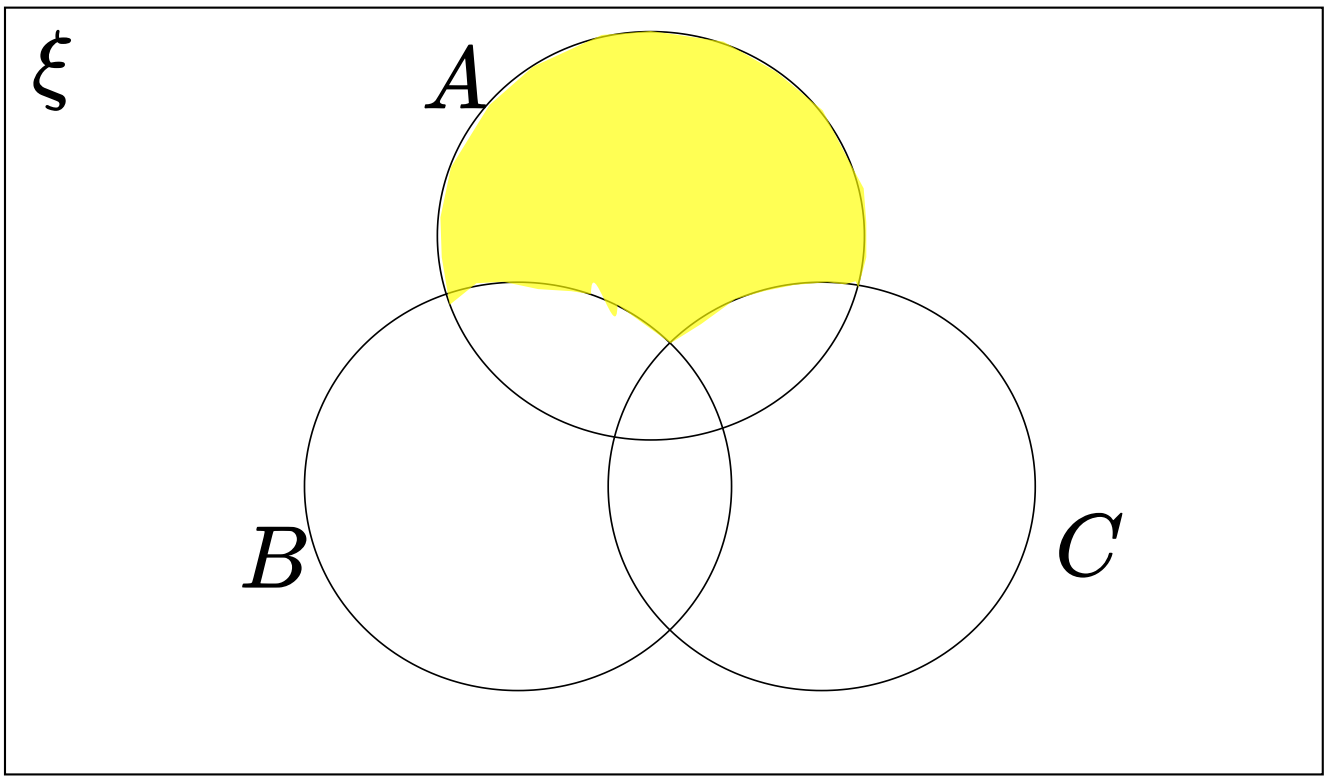




or




## What area is indicated? **Luisa · 8**



Venn Diagram can either contain:
(a) The **specific outcomes** in each set
(b) The number of items in the set (i.e. **frequencies**)
(c) The **probability** of being in that set.

This will usually be stated or made obvious from the context.

[Textbook] A card is selected at random from a pack of 52 playing cards. Let $A$ be the event that the card is an ace and $D$ the event that the card is a diamond. Find:
a) $P(A \cap D)$ b) $P(A \cup D)$ c) $P(A')$ d) $P(A' \cap D)$

Venn diagram

a

b

c

d

Think "A and D".

For union, I visualise this 'figure-of-8' shape:





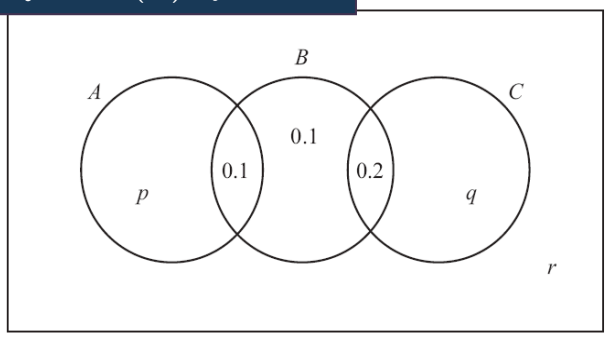
[Textbook] Given that $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cap B) = 0.25$,
a. Explain why events $A$ and $B$ are not independent.
Given also that $P(C) = 0.2$, that events $A$ and $C$ are mutually exclusive and that events $B$ and $C$ are independent,
b. Draw a Venn diagram to illustrate the events $A$, $B$ and $C$, showing the probabilities for each region.
c. Find $P((A \cap B') \cup C)$

a

b

c

## May 2013 (R) Q6




The Venn diagram in Figure 1 shows three events $A$, $B$ and $C$ and the probabilities associated with each region of $B$. The constants $p$, $q$ and $r$ each represent probabilities associated with the three separate regions outside $B$.

The events $A$ and $B$ are independent.

(a) Find the value of $p$. **(3)**



9. $P(A) = 0.55$, $P(B) = 0.35$, $P(C) = 0.4$ and $P(A \cap C) = 0.2$. Events $A, B$ are mutually exclusive; $B, C$ are independent.

(a) Draw a Venn diagram. (4)

(b) Find each probability. (1 each)

(i) $P(A' \cap B')$

(ii) $P(A \cup (B \cap C'))$

(iii) $P((A \cap C)' \cup B')$

a

b(i)

b(ii)

b(iii)

10. $P(A) = 0.25$, $P(B) = 0.4$, $P(C) = 0.45$, $P(A \cap B \cap C) = 0.1$. The pairs $A, B$ and $B, C$ are independent, and $A \cap B' \cap C = \varnothing$.

(a) Draw a Venn diagram. (4)

(b) Find (i) $P(A' \cap (B' \cup C))$, (ii) $P((A \cup B) \cap C)$. (1 each)

(c) Are $A'$ and $C$ independent? Justify. (2)

The empty set has probability zero.

a

b(i)

b(ii)

c



11. A book club reads murder mysteries $M$, ghost stories $G$ or epic fiction $E$. $P(M) = 0.5$, $P(G) = 0.4$, $P(E) = 0.6$. No member reads both ghost stories and epic fiction; $P(M \cap G) = 0.3$.

(a) Draw a Venn diagram. (4)

(b) Find (i) $P(M \cup G)$, (ii) $P((M \cap G) \cup (M \cap E))$. (2)

(c) Are $G'$ and $M$ independent? Justify. (2)

a

b(i)

b(ii)

c

12. Events $A$ and $B$ are independent with $P(A) = x$ and $P(B) = y$. Find in terms of $x, y$:

(a) $P(A \cap B)$ (2)

(b) $P(A \cup B)$ (2)

(c) $P(A \cup B')$ (2)

a

b

c

## EXTENSION (Challenge Page 20) **Luisa · 16**

Independent events $A, B, C$ have probabilities $x, y, z$. Express in terms of $x, y, z$:

(a) $P(A \cap B \cap C)$

(b) $P(A \cup B \cup C)$

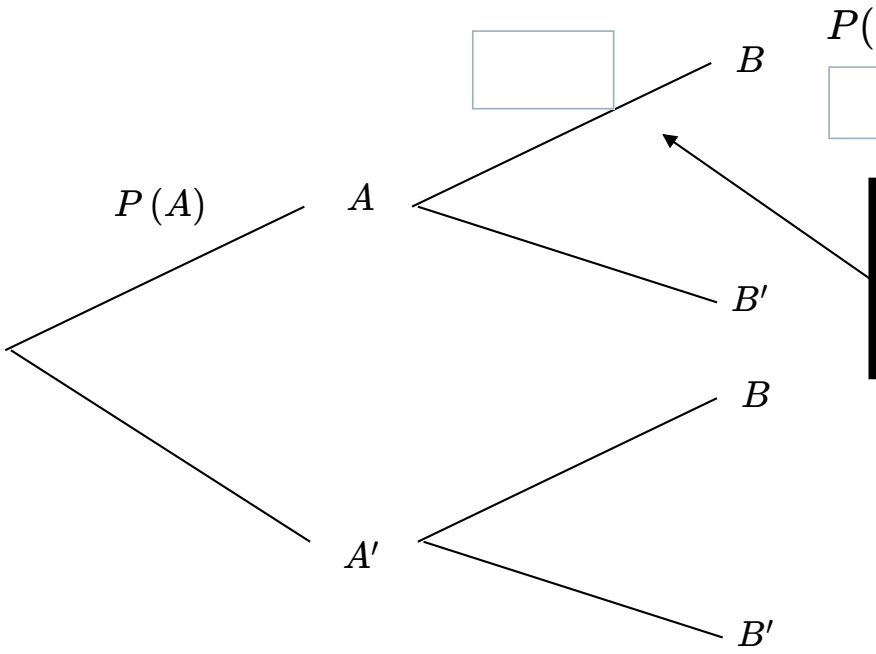
(c) $P((A \cup B') \cap C)$.

a

b

c

Think about how we formed a probability tree at GCSE:




$P(A \cap B) =$

Read the '|' symbol as "**given that**". i.e. "B occurred **given that** A occurred".

Alternatively (and more commonly):

$P(B \mid A) =$

**Memory Tip**: You're dividing by the event you're conditioning on.



1. Year 9 language choices. $B$ means the student is a boy. $F$ means the student chose French.

| | $B$ | $B'$ | Total |
|---|---|---|---|
| $F$ | 14 | 38 | 52 |
| $F'$ | 26 | 22 | 48 |
| Total | 40 | 60 | 100 |

(a) Find $P(F \mid B')$ using both methods.

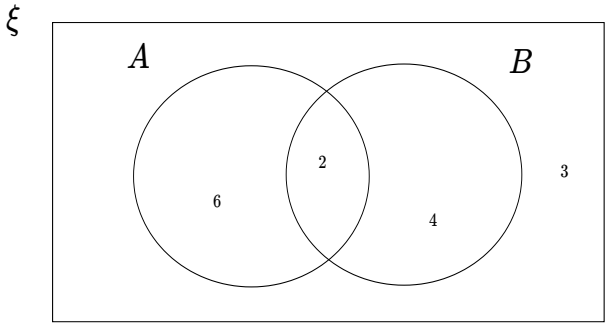
1a · Formula

1a · Restricted sample

(b) Find $P(B \mid F')$.

1b

2. Use the frequencies in the Venn diagram.




(a) Find $P(A \mid B)$ using both methods.

2a · Formula

2a · Restricted sample

(b) $P(A' \mid B')$

2b

(c) $P(B \mid A \cup B)$

2c

a Given that $P(A) = 0.5$ and $P(A \cap B) = 0.3$, what is $P(B|A)$?

a

b Given that $P(Y) = 0.6$ and $P(X \cap Y) = 0.4$, what is $P(X'|Y)$?
(Hint: Drawing a Venn Diagram will help!)

b

c Given that $P(A) = 0.5$, $P(B) = 0.5$ and $P(A \cap B) = 0.4$, what is $P(B|A')$?

c

**Note**: The 'restricted sample space' method also works for Venn Diagrams with probabilities.



The events $E$ and $F$ satisfy $P(E) = 0.28$, $P(E \cup F) = 0.76$ and $P(E \cap F') = 0.11$. Find the following probabilities.

(a) $P(E \cap F) =$ a

(b) $P(F) =$ b

(c) $P(E' \mid F') =$ c

Venn diagram

## Further Practice



1 $P(A \cap B') = 0.4, P(A \cup B) = 0.75$
Find $P(B)$, $P(A' \cap B')$.

1(i)

1(ii)

2 $P(A) = 0.47$ and $P(A \cap B) = 0.12$ and $P(A' \cap B') = 0.03$
Find $P(A \mid B')$.

2

3 $P(A') = 0.7, P(B') = 0.2, P(A \cap B') = 0.1$
Find $P(A \cup B')$, $P(B \mid A')$.

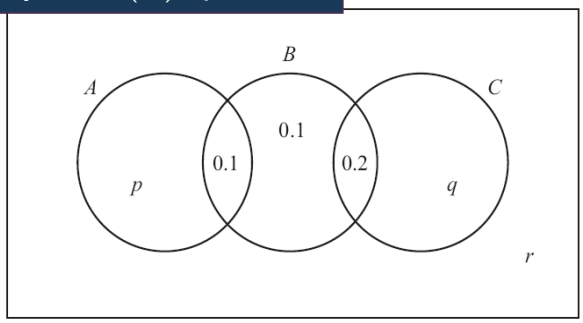
3(i)

3(ii)

## Further Test Your Understanding



### May 2013 (R) Q6




The Venn diagram in Figure 1 shows three events $A$, $B$ and $C$ and the probabilities associated with each region of $B$. The constants $p$, $q$ and $r$ each represent probabilities associated with the three separate regions outside $B$.

The events $A$ and $B$ are independent.

(a) Find the value of $p$. **(3)**

Given that $P(B|C) = \frac{5}{11}$,
(b) find the value of $q$ and the value of $r$ (4)
(c) Find $P(A \cup C|B)$ (2)

(a) **(From earlier)**
$0.1 = (p + 0.1) \times 0.4$
$p + 0.1 = 0.25$
$p = 0.15$

b

c

8. Of 120 students, 56 watch sport $S$, 77 watch drama $D$, and 18 watch both.

(a) Draw a two-way table. (2)

(b) For a randomly chosen student, find:

(i) $P(D')$ (1)

(ii) $P(S' \cap D')$ (1)

(iii) $P(S \mid D)$ (1)

(iv) $P(D' \mid S)$ (1)

a

b(i)

b(ii)

b(iii)

b(iv)



9. A rambling group has 63 women and 47 men. Of these, 26 women and 18 men use a walking stick.

(a) Draw a two-way table. (2)

(b) One rambler is selected at random. Find:

(i) $P(\text{uses a stick})$ (1)

(ii) $P(\text{uses a stick} \mid \text{female})$ (1)

(iii) $P(\text{male} \mid \text{uses a stick})$ (1)

a

b(i)

b(ii)

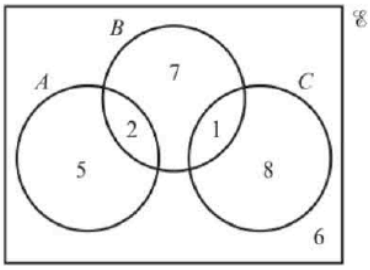
b(iii)

7. The diagram shows students who watch TV programmes $A, B, C$. A student is selected at random. Given they watch at least one programme, find the probability they watch:

(a) programme $C$ (2)

(b) exactly two programmes (2).

(c) Are watching $B$ and watching $C$ independent? Justify. (3)




a

b

c



8. $A, B$ are mutually exclusive and $B, C$ independent. $P(A) = 0.2$, $P(B) = 0.6$, $P(C) = 0.5$ and $P(A' \cap B' \cap C') = 0.1$.

(a) Draw a Venn diagram. (4)

(b) Find (i) $P(A \mid C)$, (ii) $P(B \mid C')$, (iii) $P(C \mid A \cup B)$. (1 each)

a

b(i)

b(ii)

b(iii)

9. Of 100 people, 5 have an illness. All 5 test positive, as do 10 people without the illness. Let $A$ denote having the illness and $B$ testing positive.
(a) Draw a Venn diagram. (3)
(b) Calculate $P(A \mid B)$. (2)
(c) Use b to comment on the diagnostic test. (2)

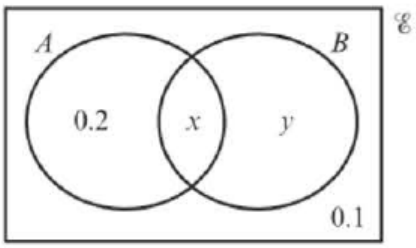
a

b

c

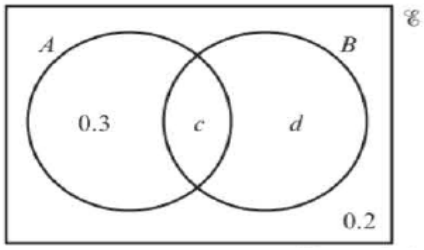


11. Given $P(A \mid B) = P(B')$, use the diagram to find $x, y$.




11

12. Given $P(A \mid B) = P(A')$, use the diagram to find $c, d$.




12

I have two children. At least one is a boy. What is the probability that both are boys? Assume independent births and equally likely sexes.



**If events A and B are independent.**

$$P(A \cap B) = P(A)P(B)$$
$$P(A \mid B) = P(A)$$

**If events A and B are mutually exclusive:**

$$P(A \cap B) = 0$$
$$P(A \cup B) = P(A) + P(B)$$

**In general:**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

We first encountered this in the previous section.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

This is known as the **Addition Law.**

**Informal Proof**: If we added the probabilities in the $A$ and $B$ sets in the Venn Diagram, we'd be double counting the intersection, so subtract so that it's only counted once.

## Example



6. Edexcel S1

(a) Explain what a sample space is. (1)

(b) Explain what an event is. (1)

$A$ and $B$ are independent with $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{4}$. Find:

(c) $P(A \cap B)$. (1)

(d) $P(A \mid B)$. (2)

(e) $P(A \cup B)$. (2)

a

b

c

d

e

## Further Examples



Textbook. $P(C) = 0.2$, $P(D) = 0.6$ and $P(C \mid D) = 0.3$. Find:

(a) $P(C \cap D)$

(b) $P(D \mid C)$

(c) $P(C \cup D)$

a

b

c

10. Jan 2012 Q2. (a) Describe the relationship between $R$ and $S$ when $P(R \cap S) = 0$. (1)

$A, B$ are independent, $P(A) = \frac{1}{4}$ and $P(A \cup B) = \frac{2}{3}$. Find:

(b) $P(B)$ (4) (c) $P(A' \cap B)$ (2)

(d) $P(B' \mid A)$ (2)

a

b

c

d

Hint for 10(b). Independence gives $P(A \cap B) = P(A)P(B)$. Knowing $P(A \cup B)$ suggests the addition rule.

Edexcel S1

9. Events $A, B, C$ are defined in sample space $S$. $A$ and $B$ are mutually exclusive. $A$ and $C$ are independent.

(a) Draw a Venn diagram showing the three events and the sample space. (3)

Given $P(A) = 0.2$, $P(B) = 0.4$ and $P(A \cup C) = 0.7$, find:

(b) $P(A \mid C)$ (2)

(c) $P(A \cup B)$ (2)

(d) $P(C)$ (4)

a

b

c

d



If I were to identify two tips that will possible help you the most in probability questions:

**If you see the words 'given that', Immediately write out the law for conditional probability.**

Example: "Given Bob walks to school, find the probability that he's not late. . ."

First thing you should write

**If you see the words 'are independent', Immediately write out the laws for independence.**
(Even before you've finished reading the question!)

Example: "$A$ is independent from $B$. . ."

First thing you should write

If you're stuck on a question where you have to find a probability given others, it's probably because you've failed to take into account that two events are independent or mutually exclusive, or you need to use the conditional probability or additional law.

9. $P(A) = 0.42$, $P(B) = 0.37$, $P(A \cap B) = 0.12$. Find:

(a) $P(A \cup B)$ (2)

(b) $P(A \mid B')$ (2)

$P(C) = 0.3$; $B, C$ are mutually exclusive and $A, C$ independent.

(c) Find $P(A \cap C)$. (2)

(d) Draw a Venn diagram showing all regions. (4)

(e) Find $P((A' \cup C)')$. (2)

a

b

c

d

e



10. $P(A) = 0.4$, $P(B) = 0.7$, $P(C) = 0.4$ and $P(A \cap B) = 0.3$. Events $A, C$ are mutually exclusive and $B, C$ independent. Find each probability. (1 each)

(a) $P(B \cap C)$

(b) $P(B \mid C)$

(c) $P(A \mid B')$

(d) $P(B \cap C \mid A')$

a

b

c

d

11. $A$ and $B$ are the events that Anna and Bella are late for school. $P(A) = 0.3$, $P(B) = 0.7$, and $P(A' \cap B') = 0.1$. Find the probability that:

(a) both are late (1)

(b) Anna is late given that Bella is late. (2)

(c) Their teacher suspects their lateness is linked. Comment and show your working. (2)

a

b

c



12. John and Kayleigh play darts on the same team. Let $J, K$ denote their respective wins. $P(J) = 0.6$, $P(K) = 0.7$ and $P(J \cup K) = 0.8$. Find the probability that:

(a) both win (1)

(b) John wins given Kayleigh loses (2)

(c) Kayleigh wins given John wins. (2)

(d) Are $J, K$ independent? Show your working. (2)

a

b

c

d

## Extension (Challenge Page 30) — Luisa · 39

$P(X = x) = kx$ for $x = 1, 2, 3, 4, 5$.

(a) Find $k$.

(b) Find $P(X = 5 \mid X > 2)$.

(c) Find the probability that $X$ is odd given that it is prime.

a

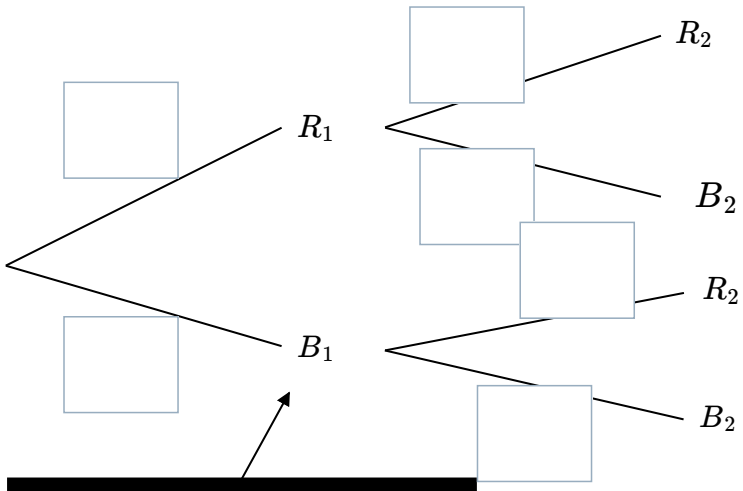
b

c

## Probability Trees — Luisa · 40

Probability Trees can be used to **determine a conditional probability.**

**Example:** You have two bags, the first with 5 red balls and 5 blue balls, and the second with 3 red balls and 6 blue balls. You first pick a ball from the first bag, and place it in the second. You then pick a ball from the second bag. Complete the tree diagram.




**Note:** Use variable subscripting to indicate what pick you're referring to.

(a) Find the probability of red on the second pick.

a

(b) Given red on the second pick, find the probability of red on the first.

b

## Further Example



Edexcel S1 May 2009 Q2. Bill travels by car, bicycle or foot with probabilities $\frac{1}{2}, \frac{1}{6}, \frac{1}{3}$. The probabilities he is late when using each method are $\frac{1}{5}, \frac{2}{5}, \frac{1}{10}$ respectively.

(a) Draw a probability tree.

(b) Given Bill is late, find the probability he did not travel on foot. (4)

a

b

## Testing Your Understanding



6. Edexcel S1, Jan 2006 Q4. A bag contains 9 blue and 3 red balls. A ball is selected at random and its colour recorded. It is not replaced. A second ball is selected at random and its colour recorded.

(a) Draw a probability tree. (3)

(b) Find the probability the second ball is red. (2)

(c) Given the second ball is red, find the probability both balls are red. (2)

a

b

c

6. Two balls are drawn at random without replacement from a bag of 4 blue and 7 green balls.
(a) Draw a probability tree. (3)
(b) Find the probability the second ball is green. (2)
(c) Given that the second ball is green, find the probability both are green. (2)

a

b

c



7. Factories A, B and C produce tin sheets. A supplies 25%, B supplies 45% and C supplies the rest. Their flaw probabilities are 0.02, 0.07 and 0.04.
(a) Draw a probability tree. (3)
(b) Find the probability a random sheet is (i) from B and flawed (2), (ii) flawed (3).
(c) Given a flawed sheet, find the probability it is from A. (3)

a

b(i)

b(ii)

c

8. A genetic condition affects 4% of a population. The probability of a positive test is 0.9 for those with the condition and 0.02 for those without it.
(a) Draw a probability tree. (3)
(b) Find the probability a random person tests negative. (3)
(c) Given a negative result, find the probability the person has the condition. (2)
(d) Comment on the test's effectiveness. (1)

a

b

c

d



9. Bill travels to work by car, bus or train with probabilities 0.1, 0.6 and 0.3. The probabilities he is late with these methods are 0.55, 0.3 and 0.05 respectively.
(a) Draw a probability tree. (3)
(b) Find the probability (i) Bill travels by train and is late (2), (ii) Bill is late (2).
(c) Given Bill is late, find the probability he did not travel by car. (4)

a

b(i)

b(ii)

c

11. A box contains 7 sweet and 3 sour jelly beans. Two are selected and eaten. Emilia wants the probability both are sweet given that at least one is sweet.

She calculates $P(\text{both sweet}) = (7/10)^2 = 49/100$ and $P(\text{at least one sweet}) = 1 - (3/10)^2 = 91/100$. Her answer is $49/91$.

Identify her mistake and find the correct probability. (4)

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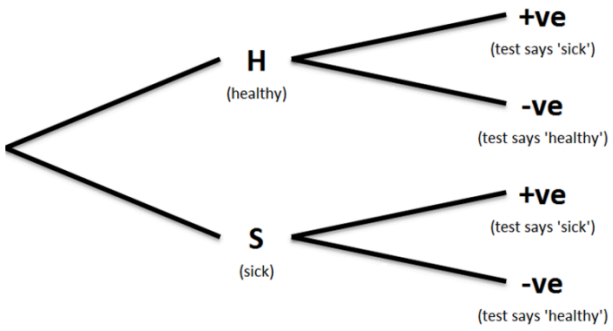


Almost all medical tests have a small chance of a 'false positive' or a 'false negative'.

- **False positive**: The test says you **do** have the condition when you really **don't**. (you're fine, but the test thinks you're sick)
- **False negative**: The test says you **don't** have the condition when you really **do**. (you're sick, but the test thinks you're fine)

- The test for a particular disease has a false positive rate of $5\%$.
  *This means that $5\%$ of people who **don't** have the disease will be told that they **do**.*
- The test has a false negative rate of $1\%$.
  *This means that $1\%$ of people who **do** have the disease will be told that they **don't**.*

Assuming this particular disease affects $10\%$of people, complete the tree diagram below:

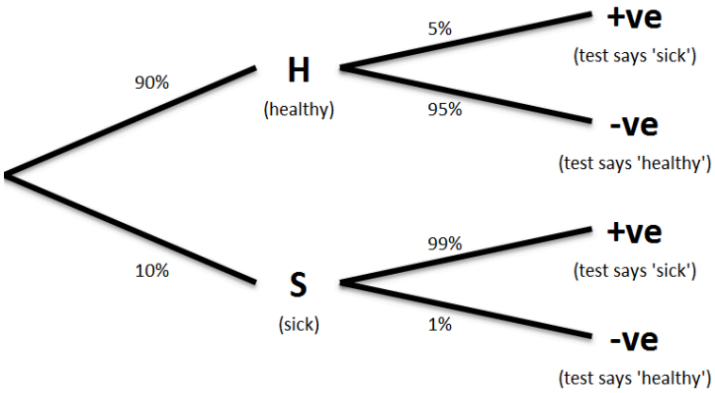


Use the probability tree to calculate the four outcomes.

| | Healthy | Sick |
|---|---|---|
| Positive | | |
| Negative | | |

A positive result for someone healthy is a false positive and can lead to unnecessary treatment or worry. A negative result for someone sick is a false negative and can delay treatment.
If the test says a person is sick, what is the probability they really are? Hint: imagine 1000 people and count the sick people among all positive tests.





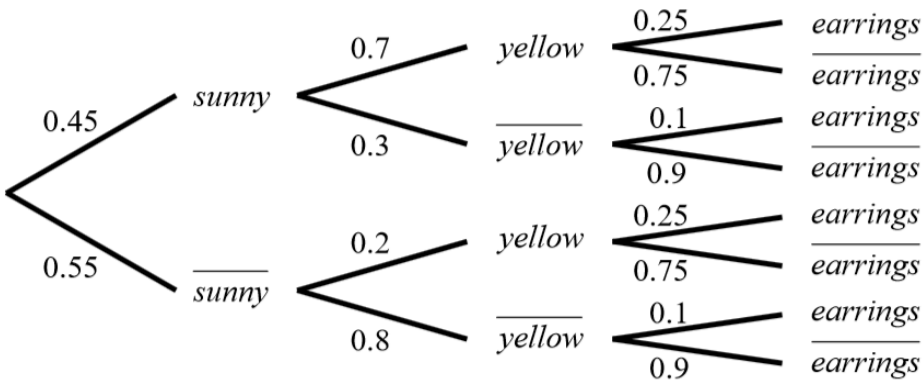

Multiply the branch probabilities.
$0.9(0.05) = 0.045$ and $0.9(0.95) = 0.855$.
$0.1(0.99) = 0.099$ and $0.1(0.01) = 0.001$.

| | Healthy | Sick |
|---|---|---|
| Positive | 4.5% | 9.9% |
| Negative | 85.5% | 0.1% |

In 1000 people, $45 + 99 = 144$ test positive. Of these, 99 really are sick.
$99/144 = 0.6875$, giving a probability of 68.75%.

Mary chooses her outfit based on the weather. The probability it is sunny is 0.45. She wears yellow with probability 0.7 when it is sunny and 0.2 otherwise. If she wears yellow, she wears her favourite earrings with probability 0.25; otherwise the probability is 0.1.
Given that Mary is not wearing her favourite earrings, find the probability she is wearing yellow.






Let $Y$ denote yellow and $E$ her favourite earrings.

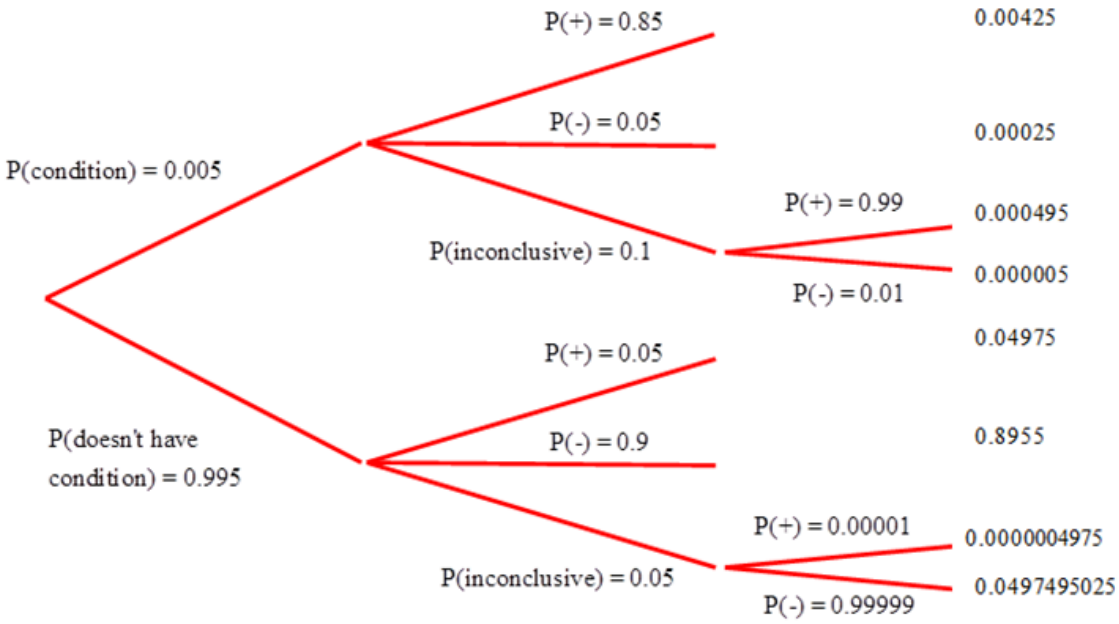
$P(Y \cap E') = 0.45(0.7)(0.75) + 0.55(0.2)(0.75) = 0.31875.$

$P(E') = 0.23625 + 0.1215 + 0.0825 + 0.396 = 0.83625.$

$P(Y \mid E') = \frac{0.31875}{0.83625} = 0.381$ (3 s.f.).

A condition affects 0.5% of a population. A test can be positive, negative or inconclusive.
Of patients with the condition, 85% test positive, 5% negative and 10% inconclusive. Of those without it, 5% test positive, 90% negative and 5% inconclusive.
Inconclusive tests are followed by a thorough examination. The examination detects the condition in 99% of patients who have it. It reports the condition in 0.001% of patients who do not have it.
Given that a patient does not have the condition, find the probability of a final positive result.






$$P(\text{positive} \mid \text{no condition}) = \frac{0.04975 + 0.0000004975}{0.995}$$
$$= 0.0500005.$$